2024 6th International Conference on Machine Learning, Big Data and Business Intelligence (MLBDBI) | 979-8-3315-4179-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/MLBDBI63974.2024.10823800

A Multiscale Hybrid Model for Natural and Remote Sensing Image-Text Retrieval

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***Abstract*—Image-text retrieval is an essential task in multimodal information processing. Recently, deep learning methods have achieved significant progress in cross-modal image-text retrieval by learning both coarse-grained representations of entire images and texts, as well as fine-grained representations of image regions and words, enabling semantic alignment at global and local levels. Drawing from multi-scale strategies often used in remote sensing image retrieval, this paper introduces multi-scale features into the framework as medium-grained image representations. We propose a model, MHL, which integrates three levels of granularity: global, local, and multi-scale. MHL extracts features from images and texts using Faster-RCNN, BERT, and a multi-scale feature fusion module, and computes three types of similarity. Our framework uses a three-stage inference approach: first, initial retrieval based on global similarity to identify candidate samples, followed by further refinement using additional similarity measures. Extensive experiments on the Flickr30K and RSICD datasets show that MHL outperforms state-of-the-art methods, demonstrating superior retrieval performance and efficiency for both natural and remote sensing images.**

Keywords-component; Cross-Modal Retrieval, Image-Text Retrieval, Feature Fusion, Remote Sensing Images, Deep Learning

I. INTRODUCTION

Images and text are fundamental carriers of information that enhance human understanding of the real world. The proliferation of multimedia devices has led to an explosive growth in the volume of images and text, which has prompted extensive research into their relationship. A foundational yet challenging topic in this field is image-text retrieval, which seeks to match content based on the semantic similarity between images and text. Most existing approaches use Convolutional Neural Networks (CNNs) to extract visual representations, while Recurrent Neural Networks (RNNs) derive textual representations, optimizing the process with a triplet loss function.

Early image-text retrieval methods [1][2][3] primarily relied on the global features of images, achieving similarity between these two modalities by directly mapping their characteristics into a shared embedding space. These coarse-grained methods focus exclusively on global matching, aligning entire images with full sentences, while neglecting the potential relationships between image regions and textual vocabulary. As a result, this approach leads to limited performance improvement and diminished interpretability.

The advancement of language representation models, such as BERT, has made fine-grained text features readily accessible. Concurrently, object detection methods, such as Faster-RCNN, have shown exceptional performance in extracting fine-grained features from images. This has resulted in growing interest in fine-grained retrieval [4][5][6][7] in recent years. Fine-grained retrieval improves performance by exploring local correspondences between visual and textual elements. Typically, attention mechanisms are used to align the components of these two modalities.

Furthermore, the rapid advancement of remote sensing image acquisition technology has significantly increased the volume of remote sensing data in recent years, resulting in numerous studies specifically focused on image-text retrieval for these images [8][9][10][12][13][14]. In contrast to natural images, remote sensing images typically exhibit multi-scale and multi-target characteristics, as well as significant background noise. To tackle these challenges, multi-scale image feature fusion techniques are frequently employed in remote sensing image processing. Input images are processed through convolutional neural networks, where convolutional features are extracted at various levels (receptive fields) and subsequently fused to obtain multi-scale integrated features. Once the fused features are obtained, they can be used for downstream tasks, such as image-text retrieval. This fusion process effectively synthesizes information across various scales, thereby enhancing the model's understanding of image details and context. Simple multi-scale image feature fusion essentially represents intermediate granularity features that exist between global and local characteristics.

Most previous research has focused either on coarse-grained representations of images and text or on fine-grained local cross-modal correspondences between image regions and text words. Recently, some studies have explored unified models that integrate both coarse-grained and fine-grained approaches. However, existing research either overlooks the characteristics of remote sensing images or is specifically designed for them, resulting in a lack of retrieval methods that address both natural and remote sensing images. To fill this gap, we propose a multi-

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scale hybrid framework, MHL (Multiscale hybrid learning framework), targeting both remote sensing and natural images. This framework employs global, local, and multi-scale features of images to achieve multi-level matching at coarse, fine, and intermediate granularities. Furthermore, based on the learned representations of features at different granularities, we introduce a fast and efficient three-stage inference method.

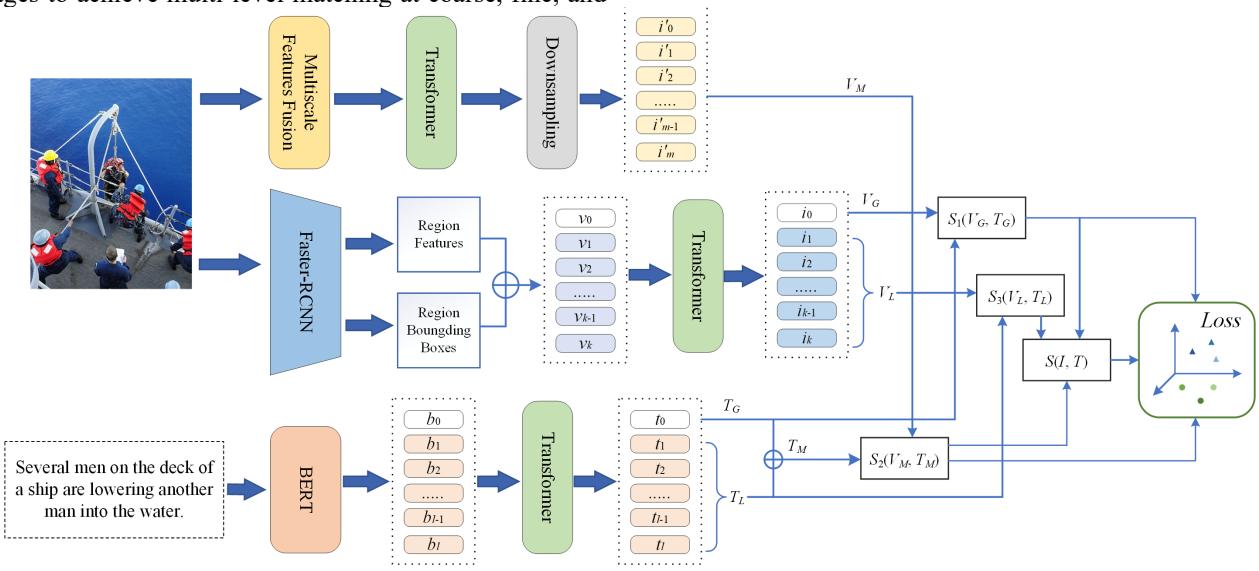


Figure 1. The architecture of the proposed MHL model.

The main contributions of this work are as follows:

- We propose an image-text retrieval framework that leverages global features, local features, and multi-scale features, effectively utilizing the advantages of different granularities in retrieval.
- We propose a three-stage inference strategy that sequentially applies global retrieval followed by local re-ranking at each stage.
- Our method achieves state-of-the-art performance on the important benchmarks, demonstrating high retrieval accuracy and efficiency.

II. METHODOLOGY

In this section, we introduce the details about our proposed model MHL. Fig.1 shows the overall architecture of the MHL. The framework comprises an image branch and a text branch. The image branch is divided into two sub-branches: one employs a Faster R-CNN and a Transformer encoder to generate global features V_G and local features V_L from the input image, while the other consists of a multi-scale fusion module, a Transformer module, and a down sampling module to produce multi-scale features V_M. The text branch employs BERT and a Transformer encoder to generate word-level features T_L and sentence-level features T_G from the input text, which are subsequently combined to form the multi-scale features T_M. The global feature similarity $S_1(V_G, T_G)$, multi-scale feature similarity $S_2(V_M, T_M)$, and local feature similarity $S_3(V_L, T_L)$ are computed, allowing the optimization of image and text features through these three similarity functions S_1, S_2, and S_3.

A. Image Presentation

Traditional image-text retrieval methods generally employ Convolutional Neural Networks (CNN) for image feature extraction. Although these methods directly map global information into a high-dimensional space, they frequently neglect local information that reflects relationships between objects and the distinction between important and redundant information, due to the complexity of the images. Solely relying on global features enhances the model's understanding of the overall image; however, it leads to inadequate fine-grained perception of the individual objects.

Many recently popular fine-grained image-text retrieval methods utilize the pre-trained Faster-RCNN network to extract local features from images. Faster-RCNN is an advanced object detector widely employed in various downstream tasks that require the extraction of salient object regions from images. As illustrated in Figure 1, we use Faster-RCNN as the image encoder to extract region features from the image. For a given image, Faster-RCNN generates a bounding box $b_i \in R^4$ and a feature vector $f_i \in R^{D'}$ for each detected object. We define the image feature representation as $V = \{v_0, v_1, \ldots v_k\}$, where $v_i = [f_i, b_i] \in R^{D'+4}$, with D' representing the dimensionality of the region features and k being the number of detected regions. v_0 denotes the global feature of the image, with the bounding box $b_0 = \{0, 0, w, h\}$, where w, and h are the width and height of the image, respectively.

To enhance the model's understanding of both global and local information, we integrate the image representations output by Faster-RCNN into a Transformer encoder module. This module consists of four identical standard Transformer encoder layers, each comprising a multi-head self-attention mechanism and fully connected feedforward operators. We denote the Transformer-based image encoder as $ITE(\cdot)$. Each element in V

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serves as a token input for the Transformer heads, producing the learned image representation as the output:

$$I = ITE(V) = \{i_0, i_1, \ldots, i_k\}, i_i \in R^D \quad (1)$$

Here, $V_G = i_0$ represents the learned global image representation, while $V_L = \{i_1, \ldots, i_k\}$ denotes the local image representations. The Transformer module projects v_i into a D-dimensional embedding i_i.

The above method achieves accurate image feature representations for natural images; however, it faces several challenges when applied to remote sensing images. In contrast to natural images, remote sensing images often contain numerous targets that can vary significantly in size within the same scene. Furthermore, remote sensing images often include extensive backgrounds and contexts unrelated to the intended focus, which can hinder the effective extraction of relevant information. Due to the high redundancy and multi-scale characteristics of remote sensing images, recent studies have employed multi-scale structures to optimize information extraction. Features obtained through multi-scale methods enhance the model's focus on salient objects while suppressing irrelevant feature representations. Therefore, our proposed model integrates multi-scale image features into the image-text retrieval framework to enhance retrieval accuracy.

Figure 1 illustrates the initial step of extracting features from the image using a multi-scale fusion module. The module utilizes four kernels of different sizes (4×4, 8×8, 16×16, and 32×32) for image subsampling, all maintaining the same stride. Each kernel generates a corresponding block that shares the same center but varies in scale. The four blocks are then projected and concatenated to create multi-scale fused image features. The self-attention mechanism of the Transformer module is then applied to enhance these feature representations, increasing their expressiveness. Finally, down sampling reduces the resolution, adjusts the number of channels for hierarchical design, and conserves computational resources. This is primarily accomplished using a convolutional operation with a stride of 2 to lower the resolution.

B. Text Presentation

Traditional text feature representation methods typically utilize RNN-based models, such as Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRU). In contrast, the pre-trained BERT model employs a self-attention-based Transformer architecture to learn semantic relationships. This model possesses robust feature extraction capabilities, generating deep bidirectional representations for word tokens. In this study, we use BERT as the text encoder. For a given sentence containing l words, BERT produces l+1 tokens representing the entire sentence and the individual words. Thus, the output of the text encoder can be expressed as $B = \{b_0, b_1, \ldots b_l\}$, where $b_i \in R^{D''}$. Here, D'' represents the dimensionality, l denotes the maximum number of words in the sentence, b_0 is the global representation of the sentence, and b_1, ..., b_l are the local representations of the l words.

Subsequently, each b_i is projected into a D-dimensional embedding through the Transformer module. Let $TTE(\cdot)$ denote the Transformer-based text encoder, which consists of four identical standard Transformer layers. Each element in B is treated as a token input to this Transformer head, producing the output as text features:

$$I = TTE(B) = \{i_0, i_1, \ldots, i_l\}, i_i \in R^D \quad (2)$$

Here, $T_G = t_0$ denotes the learned global textual representation, while $T_L = \{t_1, \ldots t_l\}$ represents the local textual representations. The global features and local features are concatenated to obtain a multi-scale fused feature representation of the text, denoted as T_M

C. Loss Function

In the framework illustrated in Figure 1, three methods are employed to calculate the similarity between two samples: 1) the global similarity $S_1(V_G, T_G)$ is computed using global features; 2) the local similarity $S_3(V_L, T_L)$ is derived from local features; and 3) the similarity $S_2(V_M, T_M)$ is calculated using the multi-scale fused image features. The final similarity S in the shared space between images and text is represented as a linear combination of the above three similarities:

$$S(I,T) = (1-\alpha-\beta)S_1(V_G, T_G) + \alpha S_2(V_M, T_M) + \beta S_3(V_L, T_L) \quad (3)$$

Here, α and β are hyperparameters used to adjust the weighting of the three similarity measures. The calculation formulas for these three cosine similarity functions are as follows:

$$S_1(V_G, T_G) = \frac{V_G \cdot T_G}{\|V_G\| \times \|T_G\|} \quad (4)$$

$$S_2(V_M, T_M) = \frac{V_M \cdot T_M}{\|V_M\| \times \|T_M\|} \quad (5)$$

$$S_3(V_L, T_L) = \frac{V_L \cdot T_L}{\|V_L\| \times \|T_L\|} \quad (6)$$

The triplet loss function is widely used in the field of image-text retrieval. We first compute the similarity score $S(I, T)$ between the image I and the text T. Subsequently, we employ the triplet loss to optimize the entire model. The loss function is defined as follows:

$$L = \sum_{\hat{I}} [\gamma - S(I,T) + S(\hat{I},T)]_+ + \sum_{\hat{T}} [\gamma - S(I,T) + S(I,\hat{T})]_+ \quad (7)$$

In this context, (I, T) represents a matching pair of image-text samples from a mini-batch, while $\hat{I}$ denotes a negative sample image and $\hat{T}$ denotes a negative sample text. The notation $[x]_+$ is equivalent to $max(x, 0)$. Based on existing research [9][13], the margin γ is set to 0.2.

During the training process, the triplet loss aims to maximize the distance between the anchor and the negative samples, while minimizing the distance between the anchor and the positive samples. Like other related studies, negative

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samples are extracted from the mini-batch for performance reasons, rather than being sourced globally.

D. Three-Stage Inference Method

This framework represents images and text using global features, local features, and multi-scale fused features. Although retrieval tasks relying solely on global features can achieve high efficiency, they often exhibit low accuracy due to insufficient understanding of local semantic information. In contrast, retrieval that utilizes local features and multi-scale fused features can attain higher accuracy; however, this approach incurs greater computational complexity, leading to reduced efficiency. To simultaneously achieve high accuracy and efficiency in retrieval, we propose a three-stage inference method.

First, a rapid retrieval method utilizing global features is employed to obtain the top $2K$ candidate samples. Next, multi-scale fused features are applied to re-rank the $2K$ candidates, resulting in the top K samples. Finally, a hybrid similarity approach that incorporates global features, local features, and multi-scale fused features is employed to re-rank these K samples, yielding the final ranking.

For example, given an input query text T_i, we retrieve K results from an image collection P containing n samples. First, we calculate the global similarity between T_i and each image I_j, denoted as $\{S_1(I_j, T_i) \mid I_j \in P\}$. Next, we select the top $2K (2K << n)$ images with the highest similarities from this set as candidate samples P_{2k}, and compute their similarities $\{S_2(I_j, T_i) \mid I_j \in P_{2k}\}$. From these, we select the top K images P_k based on the highest similarity scores. Finally, for the collection of K samples in P_k, we compute a hybrid similarity $\{S(I_j, T_i) \mid I_j \in P_k\}$ and rank the images accordingly.

III. EVALUATION

A. Dataset and Evaluation Metric

We conducted training and evaluation of the proposed method on Flickr30K [11] and RSICD [12] datasets. Flickr30K is a commonly used dataset for research on general cross-modal image-text retrieval, while RSICD is predominantly utilized for cross-modal image-text retrieval in remote sensing applications. Each image in these datasets is accompanied by five distinct descriptive sentences.

- The Flickr30K dataset comprises 31,783 images. Consistent with the methodologies outlined in [3][4], we partitioned the dataset into 29,783 images designated for training, 1,000 images for validation, and 1,000 images for testing.

- The RSICD dataset contains 10,921 remote sensing images encompassing 31 different scenes, with each image sized at 224×224 pixels. We followed the approach outlined in [9][13][14], using 80% of the data for training, 10% for validation, and 10% for testing.

We selected two evaluation metrics: Recall@K (R@K) and mean Recall (mR), which are commonly employed in cross-modal image-text retrieval tasks. Recall@K (K = 1, 5 or 10) indicates the proportion of relevant images or texts among the top K retrieval results; higher R@K values signify better performance. We conducted retrieval experiments in both image-to-text and text-to-image modes, resulting in three R@K values for each mode. The average of the six R@K values from both retrieval modes is referred to as mR.

B. Implementation Details

The experiments were performed on a workstation featuring a GeForce RTX 4090 GPU, an Intel Xeon Gold 6248R CPU, and 256 GB of RAM. To maintain consistency across various datasets and image sizes, all images were scaled uniformly to 256 × 256 pixels before inputting them into the network. The embedding dimensions for both images and text were set to 1024. During training, we used the Adam optimizer with a learning rate of 2e-4, decaying by a factor of 0.7 every 20 epochs. The training epochs and batch size were set to 70 and 100, respectively. During inference, the hyperparameters α and β in the mixed similarity were set to 0.3, while the number of samples K selected in the three-stage inference was 100. Future experiments will conduct a detailed analysis of the sensitivity of these hyperparameters. To increase the reliability of the experimental results, each group was trained and validated five times, yielding average outcomes.

C. Performance Comparison

We selected several representative and advanced image-text retrieval methods from recent years to compare with our proposed method, MHL, on the Flickr30K [11] and RSICD [12] datasets. Among the methods included in the comparison, VSE++ [3], SCAN [5], CAMP [6], and CLIP [15] provide experimental results for the two datasets. In contrast, VSRN [7], TGDT [16], and CUSA [17] present results only for the Flickr30K dataset. MTFN [18], AMFMN [8], LW-MCR [10], TGKT [13] and MSITA [14] are image-text retrieval methods specifically designed for remote sensing images and report results solely for the RSICD dataset.

Tables 1 and 2 display the experimental results on the Flickr30K and RSICD datasets. To maintain fairness, we directly reference the experimental results of each model as reported in their respective original publications. Overall, our model demonstrates the highest average retrieval accuracy on the RSICD dataset, achieving the second-highest average retrieval accuracy on the Flickr30K dataset.

As demonstrated in Tables 1, our method, MHL, achieves an mR that is approximately 20 points higher than the coarse-grained method VSE++. It also outperforms other fine-grained cross-modal retrieval methods, including SCAN, CAMP, VSRN, and TGDT. In the Flickr30K dataset, our method slightly lags behind CUSA, mainly due to CUSA's utilization of the CLIP pre-trained model and advanced semantic alignment techniques. We have not yet optimized our fine-grained semantic alignment in this area.

Tables 2 summarizes the performance of each model on the RSICD remote sensing dataset. The complexity of remote sensing images leads to significantly lower retrieval accuracy for various methods compared to results on natural image datasets. Our method achieves mR values of 24.1 on RSICD, surpassing all other methods presented in the tables. The performance of

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VSE++, SCAN, and CAMP, which are not specifically tailored for remote sensing images, is generally lower on these datasets compared to specialized remote sensing retrieval methods. Although AMFMN and MSITA consider the multi-scale nature of remote sensing images, like other methods, they focus exclusively on coarse and fine-grained features, omitting intermediate granularity, resulting in lower performance compared to our approach. In summary, our method improves cross-modal retrieval performance by integrating multi-scale information derived from both global and local features, leading to enhancements in both natural and remote sensing images.

Table 1 Comparison of text-image retrieval performance on the flickr30k dataset

Method	Text→Image			Image→Text			mR
	R@1	*R@5*	*R@10*	*R@1*	*R@5*	*R@10*	
VSE++	39.6	69.6	79.5	52.9	79.1	87.2	68.0
SCAN	48.6	77.7	85.2	67.4	90.3	95.8	77.5
CAMP	51.5	77.1	85.3	68.1	89.7	95.2	77.8
CLIP	68.7	90.6	95.2	88	98.7	99.4	90.1
VSRN	54.7	81.8	88.2	71.3	90.6	96	80.4
TGDT	66.7	92.2	97	79.6	96.9	99	88.6
CUSA	77.4	95.5	97.7	90.8	99.1	99.7	93.4
MHL	67.1	93.6	97.6	81.2	96.7	99.1	89.2

Table 2 Comparison of text-image retrieval performance on the rsicd dataset

Method	Text→Image			Image→Text			mR
	R@1	*R@5*	*R@10*	*R@1*	*R@5*	*R@10*	
VSE++	3.4	9.5	17.5	2.8	11.3	18.1	10.4
SCAN	3.7	16.4	26.7	5.9	12.9	19.8	14.2
CAMP	4.3	15.8	27.8	4.5	14.6	24.1	15.2
CLIP	5.4	20	33.2	8	21.4	32.5	20.1
MTFN	4.9	17.2	29.5	5	12.5	19.7	14.8
AMFMN	4.9	18.3	31.4	5.4	15.1	23.4	16.4
LW-MCR	4.3	18.9	32.3	4.4	13.4	20.3	15.6
TGKT	6.6	24.7	39.7	8.7	24.5	37.2	23.6
MSITA	6.1	22	35.4	8.7	22.7	33.9	21.5
MHL	7.2	26.7	38.4	10.6	25.5	36.4	24.1

IV. CONCLUSION

Image-text retrieval is challenging due to the complexity of each modality and the considerable semantic gap between them. This paper presents a cross-modal image-text retrieval framework, MHL, designed for both natural and remote sensing images. MHL extracts global, local, and multi-scale features from images and texts to enhance retrieval performance by examining multi-granular feature similarities in a shared representation space. Extensive experiments on two benchmark datasets for natural and remote sensing images show that the proposed approach outperforms state-of-the-art models in image-text retrieval tasks.

ACKNOWLEDGMENT

This work was supported by the National Defense Science and Technology Industry Foundation of China (No. JSZL2022206C001).

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